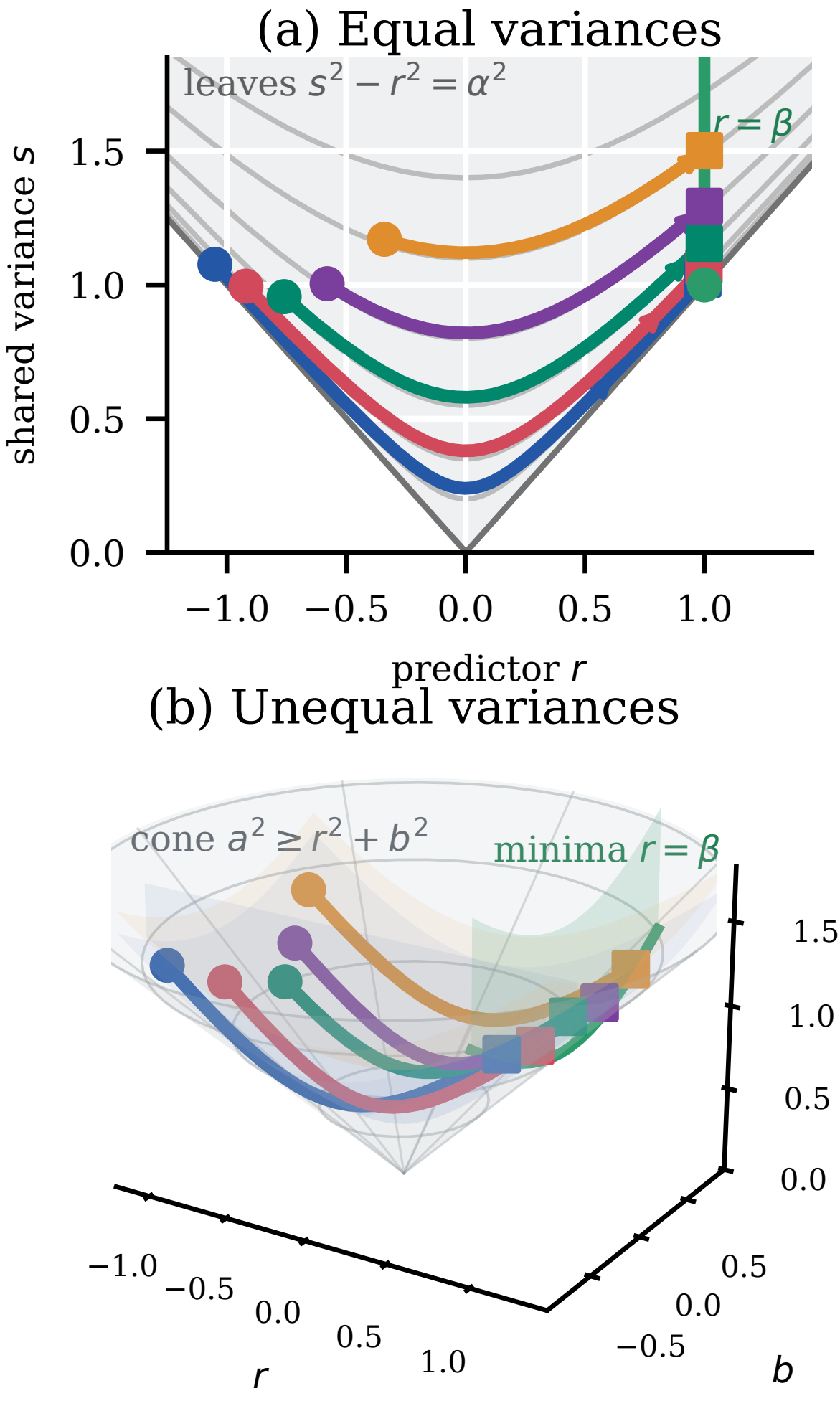

(a) Equal variances
leaves $s^2 - r^2 = \alpha^2$
$r = \beta$
shared variance s
1.5
1.0
0.5
0.0
−1.0
−0.5
0.0
0.5
1.0
predictor r
(b) Unequal variances
cone $a^2 \geq r^2 + b^2$
minima $r = \beta$
1.5
1.0
0.5
0.0
−1.0
−0.5
0.0
0.5
1.0
r
0.5
0.0
−0.5
b